TSM 098 Polynomials Test - Practice 3

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Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Multiply and simplify both products.

$(a)\quad (x+3)(x-5)$

$(b)\quad (4x-3)(5x+3)$

2. Add and simplify.

$$(2x^6-3x^5+3x^3+4x-5)+(-x^6+4x^4-2x^3+3x^2+2)$$

3. Subtract. Show distribution of the leading negative before combining terms.

$$(5x^4-2x^2+x-3)-(3x^4+4x^2-4)$$

4. Multiply and combine like terms.

$$(x^2+x-3)(2x^2-x+3)$$

5. Distribute and simplify.

$$-5x^2(2x^3-3x+3)$$

Student answer sheet

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Question 1

(a)

10 pts

(b)

10 pts

Question 2

Polynomial

20 pts

Question 3

Polynomial

20 pts

Question 4

Polynomial

20 pts

Question 5

Polynomial

20 pts

Answer key

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Question 1

(a) $x^2 - 2x - 15$

10 pts

(b) $20x^2 - 3x - 9$

10 pts

Question 2

Polynomial $x^6 - 3x^5 + 4x^4 + x^3 + 3x^2 + 4x - 3$

20 pts

Question 3

Polynomial $2x^4 - 6x^2 + x + 1$

20 pts

Question 4

Polynomial $2x^4 + x^3 - 4x^2 + 6x - 9$

20 pts

Question 5

Polynomial $-10x^5 + 15x^3 - 15x^2$

20 pts